

Question ID: 3d12b1e0

| Assessment | Test | Domain        | Skill                                                                         | Difficulty                                        |
|------------|------|---------------|-------------------------------------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Advanced Math | Nonlinear equations in one variable and systems of equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$-16x^2 - 8x + c = 0$

In the given equation,  $c$  is a constant. The equation has exactly one solution. What is the value of  $c$ ?

Correct Answer: -1

Rationale

The correct answer is  $-1$ . A quadratic equation in the form  $ax^2 + bx + c = 0$ , where  $a$ ,  $b$ , and  $c$  are constants, has exactly one solution when its discriminant,  $b^2 - 4ac$ , is equal to  $0$ . In the given equation,  $-16x^2 - 8x + c = 0$ ,  $a = -16$  and  $b = -8$ . Substituting  $-16$  for  $a$  and  $-8$  for  $b$  in  $b^2 - 4ac$  yields  $(-8)^2 - 4(-16)(c)$ , or  $64 + 64c$ . Since the given equation has exactly one solution,  $64 + 64c = 0$ . Subtracting  $64$  from both sides of this equation yields  $64c = -64$ . Dividing both sides of this equation by  $64$  yields  $c = -1$ . Therefore, the value of  $c$  is  $-1$ .